DATA ANALYTICS
PORTFOLIO CASE STUDY

Student Performance Analysis

A practical project using Excel, MySQL, Python, Pandas and Matplotlib

395
STUDENT RECORDS

10.42 / 20
AVERAGE FINAL GRADE

265
SATISFACTORY GRADES

4
ANALYSIS CHARTS

PROJECT FOCUS

Explore how student characteristics and study-related factors vary alongside final grades, then communicate the results through SQL queries, Python visualizations and an Excel dashboard.

Evidence-led documentation: screenshots in this report are the original screenshots supplied by the project author. They have been resized to fit the page; no AI-generated replacements have been used.

Dataset: UCI Student Performance — Mathematics subset (student-mat.csv)

This is a descriptive learning project. The analysis identifies patterns in this dataset and does not establish causation.

01 / OVERVIEW

Executive summary

This project examines 395 student records from the Mathematics subset of the UCI Student Performance dataset. The workflow combines spreadsheet inspection and dashboarding with SQL aggregation and Python analysis.

395 Records analysed	**10.42 / 20** Mean final grade (G3)
265 / 395 G3 ≥ 10 (67.1%)	**5.71** Mean absences

Project objectives

- Inspect the dataset and verify its structure and basic quality.
- Compare average final grades across selected student groups.
- Use SQL to aggregate and filter records.
- Reproduce key analyses in Python and create visualizations.
- Present the main findings in an Excel dashboard and document the workflow.

Workflow

EXCEL → **MYSQL** → **PYTHON** → **DASHBOARD**

Tools used: Microsoft Excel, MySQL Workbench, Python, Pandas, Matplotlib, VS Code / Jupyter Notebook.

Dataset and inspection

The project uses the Mathematics CSV file from the UCI Student Performance dataset. The working file contains 395 rows and 33 columns, including school, age, study time, past failures, absences, and grades G1, G2 and G3.

Data checks recorded during the project

- 395 student records and 33 columns in the selected CSV.
- No missing values were found in the inspected fields.
- No duplicate records were identified during the project inspection.
- Categorical values and numeric ranges were reviewed before analysis.
- The CSV uses semicolons as separators, so Python loading used the appropriate delimiter.

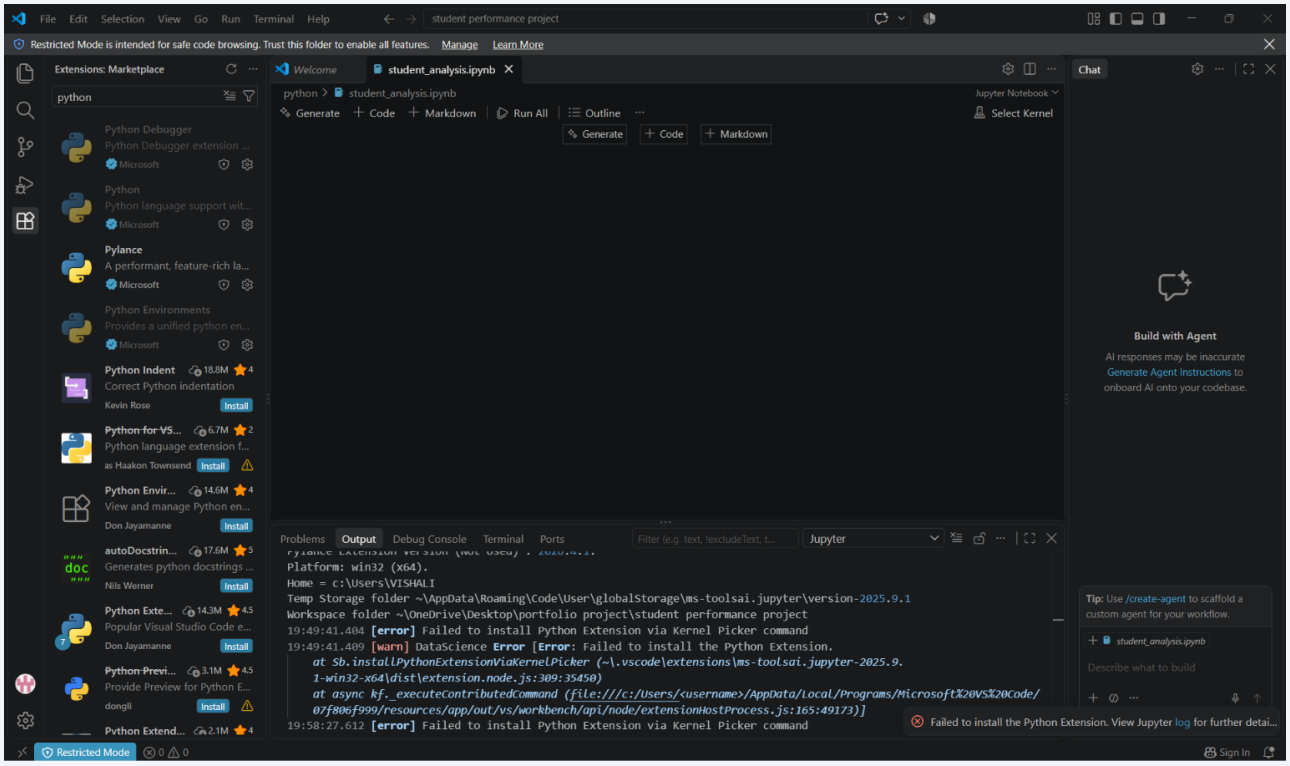


Figure 1. Original student records inspected in Excel. The screenshot is kept as supplied.

Dataset source

Cortez, P. (2008). *Student Performance* [Dataset]. UCI Machine Learning Repository. Official dataset page. DOI: 10.24432/C5TG7T.

Query-based analysis

MySQL was used to calculate grouped averages and performance percentages. These queries provide reproducible results and complement the Excel summaries.

Example analysis tasks

- Compare average G3 by school and sex using GROUP BY and AVG().
- Calculate the percentage of students with G3 ≥ 10 by sex using conditional aggregation with CASE WHEN.
- Filter and sort grouped results to compare patterns across categories.

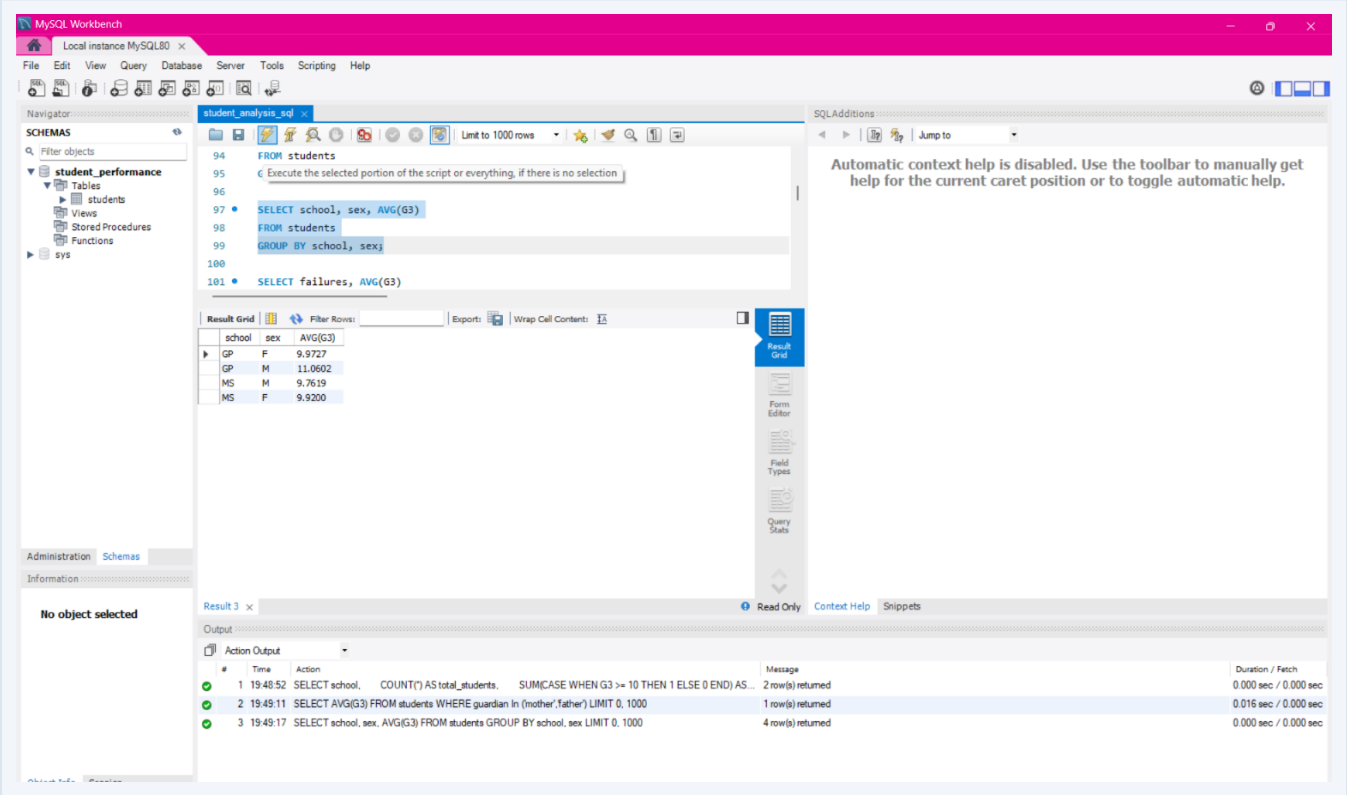


Figure 2. MySQL Workbench: average final grade grouped by school and sex.

Conditional aggregation

The second query counts records by sex and calculates the percentage meeting the project's satisfactory threshold (G3 ≥ 10). The results shown in the screenshot are approximately 70.59% for male students and 63.94% for female students in this dataset.

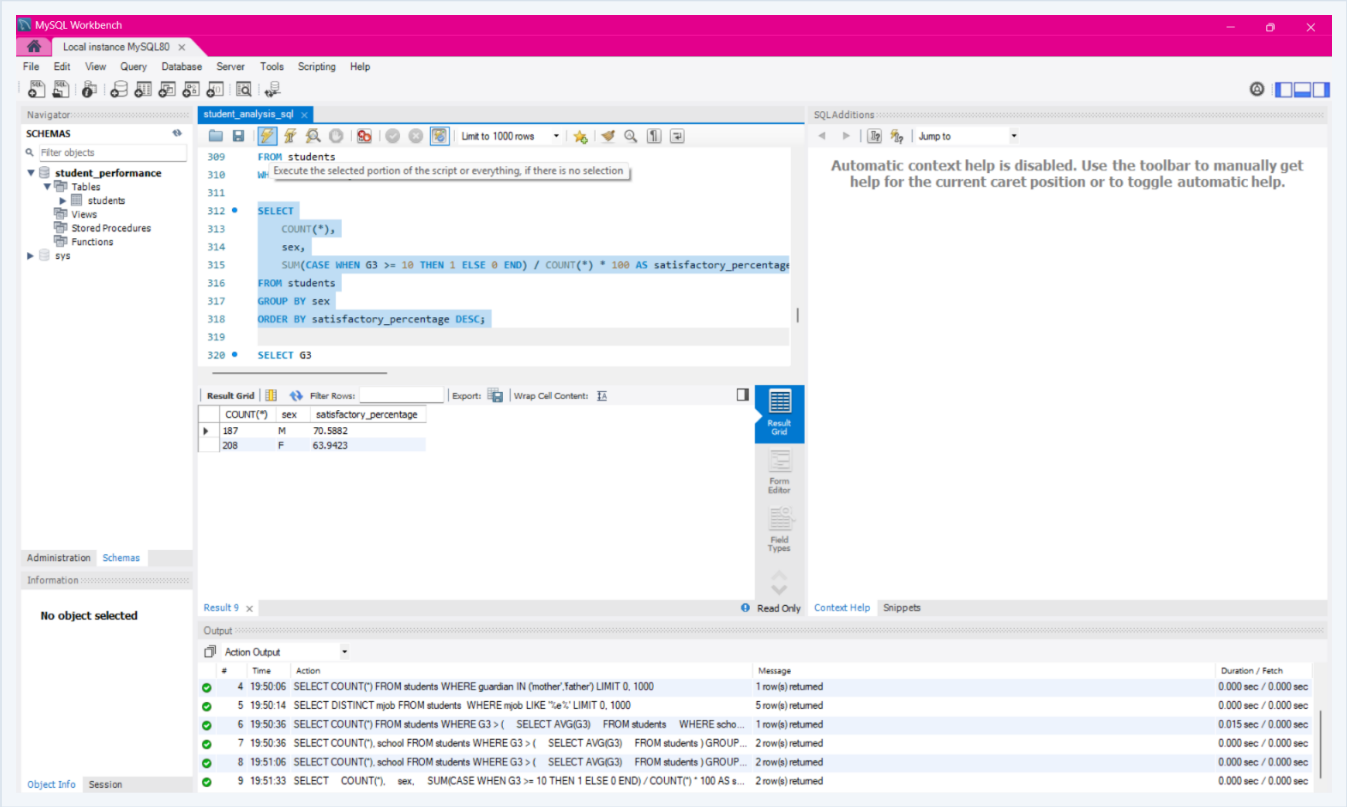


Figure 3. MySQL Workbench: satisfactory-grade percentages calculated with CASE WHEN.

Interpretation note: these are descriptive group comparisons. They do not explain why the groups differ, and should not be generalized beyond the dataset without further validation.

Analysis with Pandas

The Python notebook was used to load the semicolon-delimited CSV, inspect its shape and columns, check missing values, calculate descriptive statistics, and group average final grades by selected variables.

Examples of analyses completed

- Overall mean final grade: approximately 10.415 out of 20.
- Mean G3 grouped by school and sex.
- Mean G3 by study-time category and number of past failures.
- Grouping absences into 0–5, 6–10 and more than 10 for comparison.

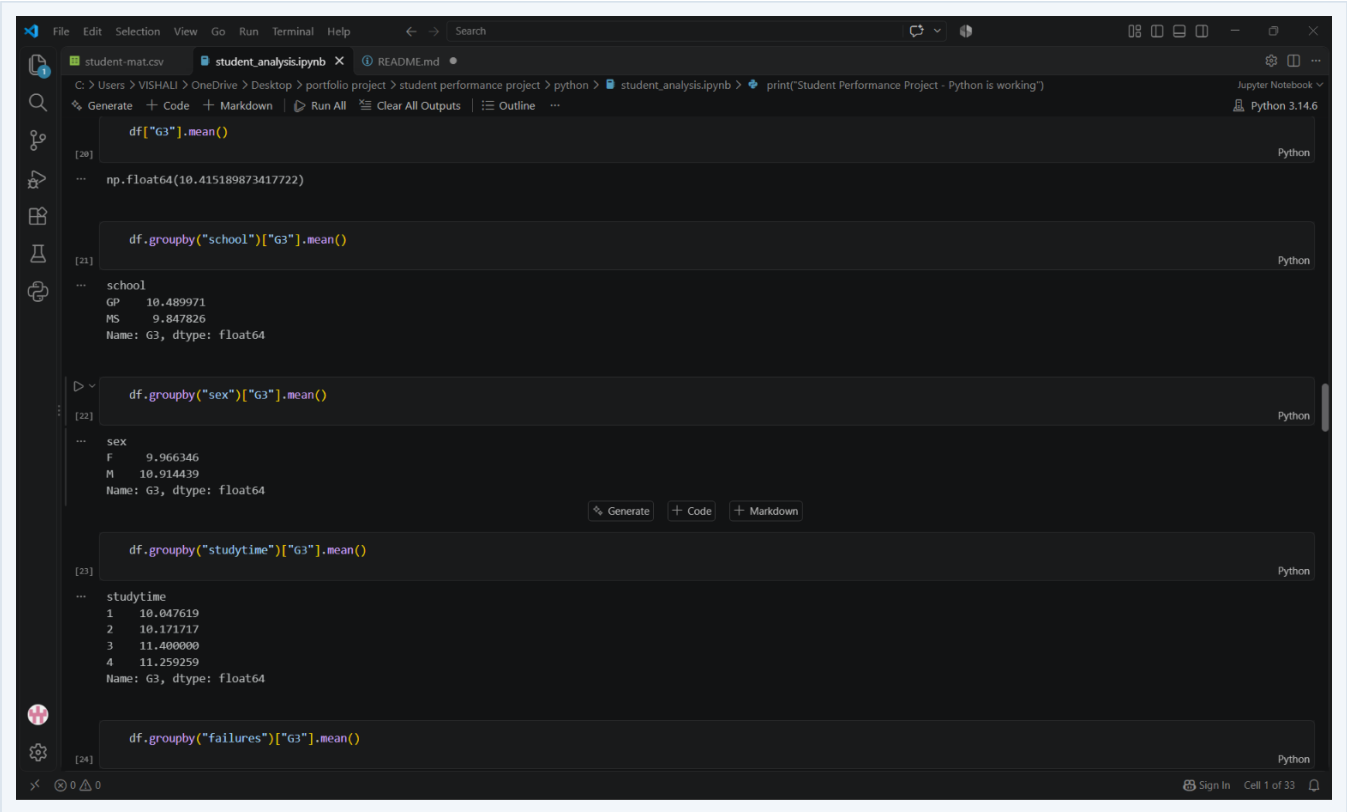


Figure 4. Original VS Code notebook output for overall grade, school, sex and study-time analysis.

Visualizing the findings

The notebook also creates charts to make the group-level patterns easier to inspect. The example below shows the average final grade decreasing as the recorded number of past failures increases.

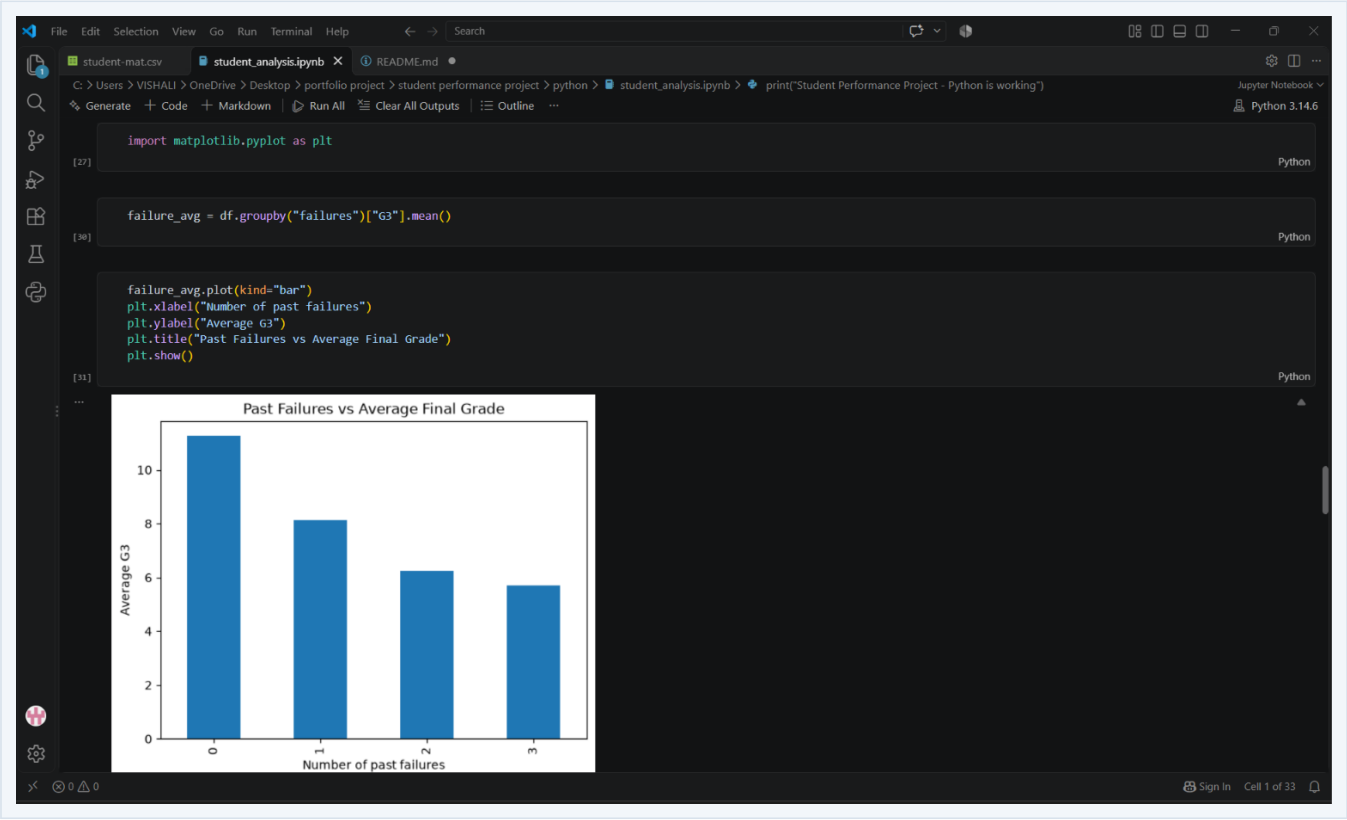


Figure 5. Original Python notebook code and generated bar chart: past failures versus average G3.

Key findings

The main results below summarize the comparisons performed in Excel, SQL and Python. Values are rounded for readability.

Analysis	Result	Careful interpretation
Overall performance	Mean G3 = 10.42/20; 265/395 (67.1%) had G3 ≥ 10.	Threshold is defined for this project; it is not a universal measure of success.
Past failures	Average G3: 11.25 (0), 8.12 (1), 6.24 (2), 5.69 (3).	A clear decreasing pattern appears across these groups in this dataset.
Study-time category	Category 3: 11.40; category 4: 11.26.	Category 3 is highest; category codes are not exact hours.
Absence group	0–5: 10.17; 6–10: 11.40; >10: 10.15.	The relationship is not a simple linear trend.
School	GP: 10.49; MS: 9.85.	The difference is modest and descriptive.
Sex	Male: 10.91; female: 9.97.	This dataset comparison does not imply a cause.

Excel dashboard

The final dashboard brings together four charts and four headline metrics to make the findings easier to review at a glance.

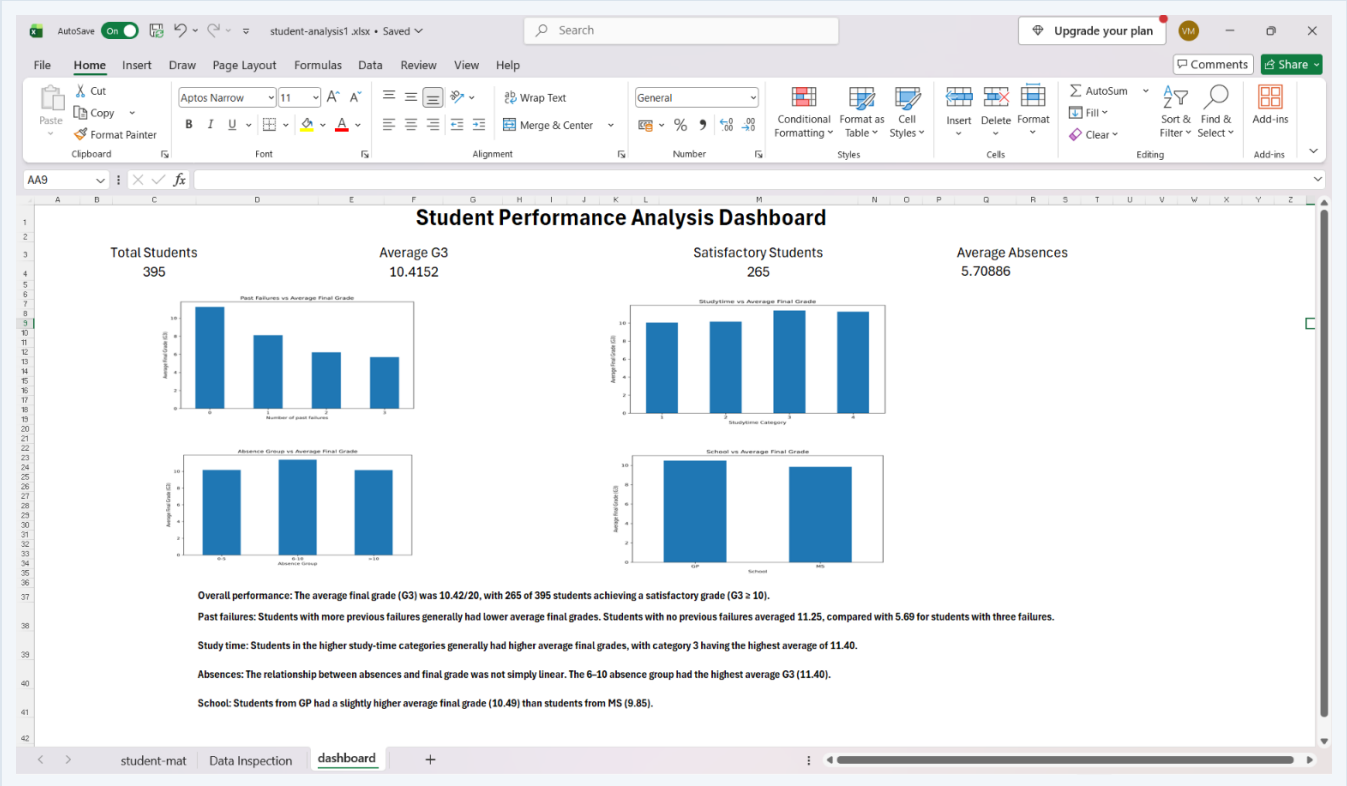


Figure 6. Completed Excel dashboard with KPI metrics, four charts and written findings.

06 / INTERPRETATION

What the results do—and do not—say

The strongest descriptive pattern is the association between a higher number of past class failures and lower average final grades. Other comparisons, such as absences and study-time categories, are more mixed.

Limitations and responsible interpretation

• The analysis is descriptive and does not establish cause and effect.
• The selected file represents the Mathematics subset from two Portuguese schools; results should not be treated as representative of all students.
• Some subgroups are small, so their averages may be less stable.
• Study-time values are coded categories, not measured hours.
• G1 and G2 are earlier-period grades and are naturally related to final grade G3; predictive work using them would need to account for that relationship.

Conclusion

This project demonstrates a practical end-to-end workflow: inspect a dataset, analyze it in Excel and SQL, reproduce comparisons in Python, visualize results, and communicate findings through a dashboard. It provides a foundation for future work such as deeper statistical testing, improved dashboard interactivity, or carefully evaluated predictive modelling.

Project deliverables

• Original dataset: student-mat.csv
• Excel workbook with inspection and dashboard
• MySQL script: student_analysis.sql
• Python notebook: student_analysis.ipynb
• Four chart image files
• This portfolio case study PDF

Portfolio status
This case study documents the work completed and the screenshots supplied. Add the public GitHub repository URL here after the repository files have been uploaded and checked.

Dataset citation: Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Source: https://archive.ics.uci.edu/dataset/320/student%2Bperformance